

UNIT CONVERSION FACTORS

Length

1 m = 100 cm = 1000 mm = $10^6 \mu\text{m}$ = 10^9 nm
1 km = 1000 m = 0.6214 mi
1 m = 3.281 ft = 39.37 in.
1 cm = 0.3937 in.
1 in. = 2.540 cm
1 ft = 30.48 cm
1 yd = 91.44 cm
1 mi = 5280 ft = 1.609 km
1 Å = 10^{-10} m = 10^{-8} cm = 10^{-1} nm
1 nautical mile = 6080 ft
1 light-year = $9.461 \times 10^{15} \text{ m}$

Area

1 cm² = 0.155 in.²
1 m² = 10^4 cm^2 = 10.76 ft²
1 in.² = 6.452 cm²
1 ft² = 144 in.² = 0.0929 m²

Volume

1 liter = 1000 cm³ = 10^{-3} m^3 = 0.03531 ft³ = 61.02 in.³
1 ft³ = 0.02832 m³ = 28.32 liters = 7.477 gallons
1 gallon = 3.788 liters

Time

1 min = 60 s
1 h = 3600 s
1 d = 86,400 s
1 y = 365.24 d = $3.156 \times 10^7 \text{ s}$

Angle

1 rad = 57.30° = $180^\circ/\pi$
1° = 0.01745 rad = $\pi/180 \text{ rad}$
1 revolution = 360° = $2\pi \text{ rad}$
1 rev/min (rpm) = 0.1047 rad/s

Speed

1 m/s = 3.281 ft/s
1 ft/s = 0.3048 m/s
1 mi/min = 60 mi/h = 88 ft/s
1 km/h = 0.2778 m/s = 0.6214 mi/h
1 mi/h = 1.466 ft/s = 0.4470 m/s = 1.609 km/h
1 furlong/fortnight = $1.662 \times 10^{-4} \text{ m/s}$

Acceleration

1 m/s² = 100 cm/s² = 3.281 ft/s²
1 cm/s² = 0.01 m/s² = 0.03281 ft/s²
1 ft/s² = 0.3048 m/s² = 30.48 cm/s²
1 mi/h · s = 1.467 ft/s²

Mass

1 kg = 10³ g = 0.0685 slug
1 g = $6.85 \times 10^{-5} \text{ slug}$
1 slug = 14.59 kg
1 u = $1.661 \times 10^{-27} \text{ kg}$
1 kg has a weight of 2.205 lb when $g = 9.80 \text{ m/s}^2$

Force

1 N = 10⁵ dyn = 0.2248 lb
1 lb = 4.448 N = $4.448 \times 10^5 \text{ dyn}$

Pressure

1 Pa = 1 N/m² = $1.450 \times 10^{-4} \text{ lb/in.}^2$ = 0.0209 lb/ft²
1 bar = 10⁵ Pa
1 lb/in.² = 6895 Pa
1 lb/ft² = 47.88 Pa
1 atm = $1.013 \times 10^5 \text{ Pa}$ = 1.013 bar
= 14.7 lb/in.² = 2117 lb/ft²
1 mm Hg = 1 torr = 133.3 Pa

Energy

1 J = 10⁷ ergs = 0.239 cal
1 cal = 4.186 J (based on 15° calorie)
1 ft · lb = 1.356 J
1 Btu = 1055 J = 252 cal = 778 ft · lb
1 eV = $1.602 \times 10^{-19} \text{ J}$
1 kWh = $3.600 \times 10^6 \text{ J}$

Mass–Energy Equivalence

1 kg ↔ $8.988 \times 10^{16} \text{ J}$
1 u ↔ 931.5 MeV
1 eV ↔ $1.074 \times 10^{-9} \text{ u}$

Power

1 W = 1 J/s
1 hp = 746 W = 550 ft · lb/s
1 Btu/h = 0.293 W